Buffers

How buffers work and the Henderson–Hasselbalch equation

Name _____ Class _____ Date _____

DATA

pKa acetic acid = 4.74 pKa NH_4^+ = 9.25 $\mathrm{pH} = \mathrm{p}K_\mathrm{a} + \log \dfrac{[\mathrm{A}^-]}{[\mathrm{HA}]}$

Give pH to two decimal places.

A. Understanding

1. What is a buffer, and what two kinds of substance must it contain?

2. Which of these mixtures are buffers? Explain each answer briefly.

(a) HCl + NaCl (b) CH_3COOH + CH_3COONa (c) NH_3 + NH_4Cl (d) NaOH + NaCl

3. Write equations to show how an acetic acid/acetate buffer removes (a) added H_3O^+ and (b) added OH^-.

B. Buffer pH

4. Calculate the pH of a buffer containing 0.10 M acetic acid and 0.15 M sodium acetate.

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5. Calculate the pH of a buffer containing 0.20 M acetic acid and 0.10 M sodium acetate.

6. Calculate the pH of a buffer containing 0.20 M ammonia and 0.10 M ammonium chloride.

C. Adding acid

7. 0.500 L of buffer contains 0.20 M acetic acid and 0.20 M sodium acetate. Calculate the pH after adding 0.020 mol of HCl.

D. Challenge

8. What ratio $[CH_3COO^-]/[CH_3COOH]$ is needed to make an acetate buffer with pH 5.00?

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